Sonali Chowdhury

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.92
St. Pauls's Boarding and Day School	Class XII (Indian School Certificate)	2019	84.50%
St. Teresa's Secondary School	Class X (Indian Certificate of Secondary Education)	2017	90.60%

EXPERIENCE

AI/ML Engineering Intern -- Blackbucks Education | June 26 - July 31 2026

- Developed an end-to-end **ML regression pipeline** using **Python, Pandas, NumPy** and **Scikit-learn** to effectively predict IPL first-innings scores from historical ball-by-ball data from 2008-2024 and performed hands on data preprocessing, feature engineering, exploratory analysis, and model comparison.
- Engineered 15+ match-state and momentum features including current score, wickets, balls remaining, run rate, innings phase and recent scoring trends and implemented time-aware preprocessing to prevent any target leakage and compared **Linear Regression, Random Forest, Gradient Boosting and XGBoost,** selecting **XGBoost** after chronological validation, achieved 11.8 MAE, 15.6 RMSE and 0.86 R^2 on the 2024 hold-out season.
- Built a reproducible **Scikit-learn Pipeline** with categorical encoding and Joblib-based model serialization, and developed a Streamlit interface for real-time predictions with input validation and uncertainty intervals.

'LLMs From Scratch' Summer Workshop -- Headed by Prof. Anurag Tripathi | 07 May - July 30 2026

- Learned the fundamentals of building Large Language Models (LLMs) from first principles.
- Explored Transformer architecture, tokenization, embeddings, self-attention, positional encoding and LLM concepts.

SKILLS

- **Languages** : Python, SQL, NumPy, Pandas, Skicit-Learn, PyTorch.
- **Statistics** : Descriptive Statistics, Probability, Probability Density Functions, Hypothesis Testing, A/B Testing, ANOVA.
- **AI/ML** : Machine learning (supervised and unsupervised), Linear and Logistic Regression, KNN, Decision Trees, Random Forest, SVM, PCA, Hyperparameter Tuning, Natural Language Processing and Time-Series Modelling.
- **Deep Learning & GenAI** : Transformers, RAG pipelines, LLMs.

PROJECTS

Project 1: CerviCell Research Copilot -

- Built a research assistant using **NLP, RAG, semantic search, and LLM-based research workflows** for querying cervical cancer and cytology literature, by developing an end-to-end document pipeline for **PDF extraction, recursive chunking, Sentence Transformer embedding** (all-MiniLM-L6-v2)**, and ChromaDB.**
- Implemented **semantic retrieval, relevance filtering, and evidence extraction** to generate responses and reject queries outside the research corpus, and implemented a **Transformer language model**, incorporating multi-head self-attention, causal masking, positional embeddings, residual connections, and feed-forward networks.
- Explored **LLM-based answer generation** using an instruction-tuned model and compared it with evidence-based generation to reduce unsupported biomedical responses, structuring the project into a modular, reproducible **ML/NLP pipeline** covering document ingestion, preprocessing, embeddings, vector retrieval, Transformer training, and inference.

Project 2: Healthcare Query Response System -

- Built an end-to-end healthcare NLP system using **14,845 MedQuAD question-answer pairs**, mapping user queries into healthcare intents such as symptoms, treatment, diagnosis, causes, and prognosis.
- Compared **Logistic Regression, Random Forest, XGBoost,** and a **custom Transformer encoder**; Logistic

Regression model achieved **54.17% accuracy on 48 manually authored unseen queries**, providing a more realistic measure of generalization than the near-perfect benchmark scores on the original dataset.
- Integrated the classifier with an **LLM-based response generation layer** and built **FastAPI REST endpoints** for prediction and model comparison, creating a pipeline from user query to intent classification to generated response.

Project 3: Multimodal Sepsis Prediction Using Uncertainty-Guided Adaptive Fusion -

- Worked on a multimodal sepsis prediction framework using the MIMIC-III ICU dataset, combining structured patient vitals with unstructured clinical notes for getting important clinical insights.
- Performed data preprocessing and cohort selection, including normalization, missing-value handling, clinical-text cleaning, tokenization, and temporal alignment of patient vitals and clinical notes, developing a structured-data branch using LSTM to capture temporal dependencies and identify physiological trends for improved prediction.
- Applied BERT with Attention to clinical notes to extract contextual and semantic information from complex medical text and integrated structured and textual representations through an uncertainty-guided adaptive fusion layer.
- Incorporated evidential deep learning with Dirichlet-based uncertainty estimation to generate prediction probabilities and uncertainty scores, enabling risk stratification into **low, medium and high**-risk cases.
- **Results** - Accuracy: 59.45%, Precision: 58.23%, Recall: 79.31%, F1 Score: 67.15%, ROC-AUC: 71.5%

EXTRACURRICULAR

Poetry writing -

- featured poem in the 6th edition of Pulse Magazine, published by the Sunshine Cell at IIT Hyderabad.
- Participated in a poetry writing competition 'Sunday Poetry Sprint', conducted by the writing club at IIT Hyderabad.
- Published a poem in a paperback magazine named 'Transcendental' available on Amazon.

Other activities include playing Sudoku and learning ukulele as personal hobbies for catharsis.